

M02.A02: Model-Based - DT, & k-Map

This week's assignment is an individual assignment with two parts. Please be sure to submit both parts.

Part I [40%]:

Do Problem 8 on page 131 of Jorgensen's Software Testing.

"The retirement pension salary of a Michigan public school teacher is a percentage of the average of their last 3 years of teaching. Normally, the number of years of teaching service is the percentage multiplier. To encourage senior teachers to retire early, the Michigan legislature enacted the following incentive in May of 2010; Teachers must apply for the incentive before June 11, 2010. Teachers who are currently eligible to retire (age ≥ 63 years) shall have a multiplier of 1.6% on their salary up to, and including, \$90,000, and 1.5% on compensation in excess of \$90,000. Teachers who meet the 80 total years of age plus years of teaching shall have a multiplier of 1.55% on their salary up to, and including, \$90,000 and 1.5% on compensation in excess of \$90,000.

Create a Decision Table to describe the retirement pension policy requirements; be sure to consider the retirement eligibility criteria carefully.

What is the compensation multiplier for a person who is currently 64 with 20 years of teaching whose salary is \$95,000?

Comment on the completeness and vagueness of the requirements. If possible, calculate the default rules. "

Be sure to include your assumptions and complete decision table plus any reductions that simplify the table to reach your final answer.

Part II [60%]:

You are required to test software filter for the selection of students into the basketball team of the university. The basketball league wouldn't allow players to represent their university unless each player has at least 12 credits per term within the university. The university coach is interested only in players who weigh more than 180 pounds and are at least 77 inches tall.

You are required to:

- 1) Identify the input conditions and corresponding actions.
- 2) Provide the initial Decision Table, reduce it and show the reductions steps.
- 3) Comment on the completeness and vagueness of the requirements.
- 4) If possible, calculate the default rules.

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Part I: Retirement Pension Decision Table

Assumptions

- The application deadline (“before June 11, 2010”) is treated as its own condition (C1), since none of the incentive's benefits apply unless a teacher applies in time.
- If a teacher applies on time but satisfies neither eligibility (age ≥ 63 nor the Rule of 80), the incentive does not apply and the teacher's pension reverts to the normal formula given in the problem's introduction. This normal formula is used as the Default Action (A4).
- The statute does not say which multiplier applies when a teacher satisfies BOTH eligibility at once (age ≥ 63 AND Rule of 80). This is treated as an ambiguity in the requirements. The assumption made here is that the higher rate to the retiree applies in that overlap case since the statute gives no explicit precedence and pension provisions are conventionally read in favor of the beneficiary absent contrary language. This assumed rule is marked with a * in the tables below.
- Age and years of teaching are assumed to be evaluated as of the application date.

Step 1 & 2: Conditions and Actions

- C1: Applied for the incentive before June 11, 2010? (Y/N)
- C2: Is the teacher currently eligible to retire, i.e. age ≥ 63 ? (Y/N)
- C3: Does the teacher meet the Rule of 80 (age + years of teaching ≥ 80)? (Y/N)
- A1: Multiplier on the first \$90,000 of salary = 1.6%
- A2: Multiplier on the first \$90,000 of salary = 1.55%
- A3: Multiplier on the portion of salary in excess of \$90,000 = 1.5%
- A4: Default action, no incentive applies; use the normal pension formula (years of teaching service is the multiplier)

Step 3: Initial Decision Table ($2^3 = 8$ rules)

	Values	1	2	3	4	5	6	7	8	Total
CONDITIONS										
C1: Applied before Jun 11, 2010?	Y/N	Y	Y	Y	Y	N	N	N	N	
C2: Age $\geq$ 63?	Y/N	Y	Y	N	N	Y	Y	N	N	
C3: Age + Years $\geq$ 80 (Rule of 80)?	Y/N	Y	N	Y	N	Y	N	Y	N	
ACTIONS										

A1: 1.6% on first \$90,000		Y*	Y							
A2: 1.55% on first \$90,000				Y						
A3: 1.5% on amount over \$90,000		Y	Y	Y						
A4: Default — normal formula (years of teaching = multiplier)					Y	Y	Y	Y	Y	
Checksum		1	1	1	1	1	1	1	1	8

* Rule 1 (C1=Y, C2=Y, C3=Y) is the ambiguous overlap case; A1 is assigned here per the stated assumption.

Step 4: Reductions

Rules 5–8 (C1 = N) all yield the same action, A4, regardless of C2 or C3, since a teacher who did not apply on time gets the default formula no matter their age or years of service. These four columns collapse into a single rule with C2 and C3 marked “X” (don’t care).

Rules 1 and 2 (C1=Y, C2=Y) both yield A1 + A3 regardless of C3, per the stated assumption that the age-based rate governs whenever a teacher qualifies by age, whether or not the Rule of 80 also applies. These two columns collapse into a single rule with C3 marked “X.”

Rules 3 (C1=Y, C2=N, C3=Y) and 4 (C1=Y, C2=N, C3=N) cannot be merged with each other or with any other rule, since each has a different action and set of true predicate values.

Final Reduced Decision Table (4 rules)

	Values	R1	R2	R3	R4	Total
CONDITIONS						
C1: Applied before Jun 11, 2010?	Y/N	Y	Y	Y	N	
C2: Age ≥ 63?	Y/N	Y	N	N	X	
C3: Age + Years ≥ 80 (Rule of 80)?	Y/N	X	Y	N	X	
ACTIONS						
A1: 1.6% on first \$90,000		Y*				

A2: 1.55% on first \$90,000			Y			
A3: 1.5% on amount over \$90,000		Y	Y			
A4: Default — normal formula				Y	Y	
Checksum		2	1	1	4	8

Checksum verification: $2 + 1 + 1 + 4 = 8$, matching the initial table's total of 8. No rules were lost in the reduction.

Default Rules

Of the 8 total combinations, only 2 ($C1=Y, C2=Y, C3=N$ and $C1=Y, C2=N, C3=Y$) are explicitly and unambiguously described by the statute's text. The remaining 6 combinations are not directly addressed by the incentive language and fall back on the Default Rule/Action, A4 (the normal pension formula):

- $C1=N$ (4 combinations): teacher did not apply before the deadline, regardless of age or years of service; default applies.
- $C1=Y, C2=N, C3=N$ (1 combination): teacher applied on time but meets neither eligibility path; default applies.
- $C1=Y, C2=Y, C3=Y$ (1 combination): the ambiguous overlap case. This one is not truly covered by the default rule; it requires an explicit assumption (see above) rather than falling back to the normal formula, since the teacher clearly does qualify for some incentive multiplier; the statute just doesn't say which.

So, of the 6 unaddressed combinations, 5 are resolved by the Default Rule (A4), and 1 (the overlap case) required a separate disambiguating assumption rather than a default fallback.

Compensation Multiplier: Age 64, 20 Years of Teaching, Salary \$95,000

Age ≥ 63 ? ; Yes ($64 \geq 63$) $\rightarrow C2 = Y$.

Rule of 80? Age + years of teaching = $64 + 20 = 84 \geq 80 \rightarrow C3 = Y$.

Assuming the application was submitted before the June 11, 2010 deadline ($C1 = Y$), this teacher satisfies BOTH eligibility paths, the ambiguous overlap case (Reduced Table Rule R1).

Per the stated assumption (the higher, more advantageous rate governs), the

applicable multiplier is: 1.6% on the first \$90,000 of salary, and 1.5% on the remaining \$5,000 (the portion of the \$95,000 salary above \$90,000).

$(0.016 \times \$90,000) + (0.015 \times \$5,000) = \$1,440 + \$75 = \mathbf{\$1,515}$ in annual pension credit from this tiered multiplier, applied against the average of the teacher's last 3 years of salary

Completeness and Vagueness of the Requirements

The requirements leave out a few important details. The biggest gap is that they never say what multiplier to use if a teacher qualifies both ways at once (age 63+ and the Rule of 80), I had to decide to pick the higher rate. It's also not completely clear if applying exactly on June 11 counts as "before" the deadline or not, and the rules never say what date to use when checking someone's age or years of teaching. On top of that, it's not clear whether the \$90,000 cutoff is based on the same 3-year salary average used for the base pension, or just their current salary. None of these are major issues, but small wording gaps that could easily lead to two people calculating someone's pension differently.

Part II: Basketball Team Selection Filter

1) Input Conditions and Actions

Input Conditions:

- C1: Credits per term ≥ 12 ? (Y/N)
- C2: Weight > 180 pounds? (Y/N)
- C3: Height ≥ 77 inches? (Y/N)

Actions:

- A1: Player is selected for the team
- A2: Player is rejected $\rightarrow$ does not meet the league's credit requirement
- A3: Player is rejected $\rightarrow$ meets the credit requirement, but does not meet the coach's weight/height criteria

2) Initial Decision Table ($2^3 = 8$ rules)

	Values	1	2	3	4	5	6	7	8	Total
CONDITIONS										
C1: Credits ≥ 12?	Y/N	Y	Y	Y	Y	N	N	N	N	
C2: Weight > 180 lb?	Y/N	Y	Y	N	N	Y	Y	N	N	
C3: Height ≥ 77 in?	Y/N	Y	N	Y	N	Y	N	Y	N	
ACTIONS										
A1: Selected for the team		Y								
A2: Rejected — insufficient credits (ineligible per league)						Y	Y	Y	Y	
A3: Rejected — does not meet coach's physical criteria			Y	Y	Y					
Checksum		1	1	1	1	1	1	1	1	8

Reduction Steps

Step 1: Rules 5–8 ($C1 = N$) all yield A2 regardless of C2 or C3 — a player without enough credits is rejected no matter their weight or height. These four columns collapse into a

single rule with C2 and C3 marked “X.”

Step 2: Among the remaining rules with C1 = Y, rules 2 (Y,Y,N) and 4 (Y,N,N) both have C3 = N and both yield A3; they differ only in C2, so they merge into a single rule with C2 marked “X.” Rule 3 (Y,N,Y) still yields A3 but cannot be merged further, since it differs from the other A3 rules in more than one condition at a time.

Rule 1 (Y,Y,Y → A1, selected) is unique and cannot be merged with any other rule, since it is the only rule yielding A1.

Final Reduced Decision Table (4 rules)

	Values	R1	R2	R3	R4	Total
CONDITIONS						
C1: Credits ≥ 12?	Y/N	Y	Y	Y	N	
C2: Weight > 180 lb?	Y/N	Y	X	N	X	

	Values	R1	R2	R3	R4	Total
C3: Height ≥ 77 in?	Y/N	Y	N	Y	X	
ACTIONS						
A1: Selected for the team		Y				
A2: Rejected — insufficient credits					Y	
A3: Rejected — does not meet physical criteria			Y	Y		
Checksum		1	2	1	4	8

Checksum verification: $1 + 2 + 1 + 4 = 8$, matching the initial table's total of 8. No rules were lost in the reduction.

No further reduction is possible: R2 and R3 cannot be combined because they differ in two conditions at once (C2 and C3), and combining them into a single “X, X” rule would incorrectly also absorb R1's case (Y, Y, Y).

3) Completeness and Vagueness of the Requirements

The requirements are simple but leave a few things unclear. They never say whether "credits per term" means every single term, just the most recent one, or an average. There's also no

upper limit given for credits, weight, or height, just minimums. Not to mention, there's a small inconsistency worth noting: weight has to be strictly "more than" 180 pounds, so exactly 180 doesn't count, but height only has to be at least 77 inches, so exactly 77 does count, conditions that are easy to miss. Finally, the rules don't explicitly say all three conditions need to be true at the same time to make the team; I assumed that's the intent, but it's technically not stated. Real eligibility rules would likely also care about GPA or academic standing, which isn't mentioned here at all.

4) Default Rules

This specification is logically complete: all $2^3 = 8$ possible combinations of the three binary conditions are explicitly and unambiguously covered by the stated rules (every combination results in exactly one of A1, A2, or A3, with no combination left undefined). The initial table's checksum of 8 matches the maximum possible number of rules for 3 binary conditions, so no Default Rule is required for this part.

"I pledge my honor that I have abided by the Stevens Honor System." SK